

DURANJAI VENKATACHALAPATHI

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Summary

Results-driven AI Engineer focused on bridging the gap between cutting-edge machine learning research and scalable product development. Expert in architecting end-to-end intelligent systems, specializing in Large Language Model (LLM) integration, Retrieval-Augmented Generation (RAG) pipelines, and real-time computer vision applications. Committed to building robust, evaluable AI solutions that prioritize accuracy, performance, and seamless user experiences.

Sep 2023

Education

Easwari Engineering College

Sep 2023 – Apr 2027

B.E in Computer Science & Engineering, Artificial Intelligence & Machine Learning — CGPA: 8.71

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Internships

OEG Defence & Aerospace Pvt. Ltd.

June 2025– Present

Machine Learning Intern

India

- Contributed to the development of intelligent drone systems by integrating machine learning and computer vision techniques
- Developed an **Autonomous Drone Person Tracking** system using YOLO object detection with PID control algorithms and live GPS update rate management ([GitHub](#))
- Assisted in creating a drone-based object detection system using YOLOv8 for real-time target identification
- Supported dataset preparation, annotation, and augmentation workflows through RoboFlow
- Collaborated on Python-based model deployment including real-time camera inference, OpenCV integration, and model fine-tuning

Projects

Weather Agent | LLM-Powered Tool-Using Agent

2025

- Built a ReAct-style AI agent using Llama-4 LLM with Open-Meteo API integration for real-time weather forecasting, featuring geocoding, strict tool-calling rules, and CLI interface ([GitHub](#))

CoursePack-QA | RAG System for Student Q&A

2025

- Developed an end-to-end Retrieval-Augmented Generation (RAG) system to answer student questions from course documents
- Implemented hybrid search combining vector embeddings and BM25 for improved retrieval accuracy
- Integrated reranking pipeline to provide grounded, citable answers from course materials

Barista Video Analysis | Local Computer Vision & Vision-Language Application

2025

- Developed a local computer vision system to analyze barista POV videos and identify actions, drink types, visible ingredients, and preparation stages ([GitHub](#))
- Implemented frame sampling and video processing pipelines using OpenCV and yt-dlp
- Integrated a local vision-language model (Moondream via Ollama) for fully offline inference without paid APIs
- Designed an event-driven workflow using LlamaIndex Workflows with structured result aggregation and timeline analysis
- Built an interactive Gradio web interface for visualizing frame-by-frame insights and ingredient frequency

Technical Skills

AI & Machine Learning: RAG, Hybrid Search (Vector + BM25), Reranking, Computer Vision, LLMs, Prompt Engineering, Vector Embeddings (BGE, MiniLM), Cross-Encoders, RAGAS

Languages: Python, SQL

Data Libraries: Pandas, NumPy

Databases: Vector Databases (pgvector, Redis), SQLite

MLOps & Tools: Git, GitHub, Docker, Streamlit

Web: HTML, CSS

Spoken Languages: English, Tamil

Soft Skills: Communication, Team Collaboration, Problem Solving, Leadership, Critical Thinking